

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 2

Course Code: CSE252

Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Course Outcome: CO2

Paper: Sample Multiple Choice Questions - ANSWER KEY

Maximum Marks: 40 x 1 = 40

Unit: 2

Unit Title: Data Preparation and Machine Learning Workflow

Unit Mapped RBT: L3 (Apply)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the Pandas structure that holds a single labelled column of data.

- A) A Series <-- correct
- B) A DataFrame
- C) An ndarray
- D) A dictionary

Answer: A. A Series is one labelled column; a DataFrame is a table made of several Series sharing one index.

Q2. Identify the Pandas method that returns count, mean, standard deviation and quartiles for the numeric columns.

- A) info()
- B) describe() <-- correct
- C) head()
- D) shape

Answer: B. describe() gives the numeric summary, while info() reports the data types, the non-null counts and the memory used.

Q3. State the Matplotlib plot used to show the distribution of a single numeric column.

- A) A scatter plot
- B) A line plot
- C) A histogram <-- correct

D) A pie chart

Answer: C. A histogram bins the values and plots the counts, which reveals skew, spread and any second peak in the column.

Q4. Name the kind of dataset made up of photographs and free text.

- A) Structured data
- B) Tabular data
- C) Relational data
- D) Unstructured data <-- correct

Answer: D. Unstructured data has no fixed columns. Structured data sits in rows and columns; semi-structured data such as JSON carries its own tags.

Q5. Identify the scaler that uses the median and the interquartile range rather than the mean and standard deviation.

- A) The standard scaler
- B) The robust scaler <-- correct
- C) The min-max scaler
- D) The normaliser

Answer: B. Median and interquartile range are resistant to extreme values, so the robust scaler suits a column that holds outliers worth keeping.

Q6. State what a validation partition is used for.

- A) Fitting the parameters of the final model
- B) Reporting the final performance of the system

- C) Storing the records that have missing values
- D) Choosing between models and settings before the test set is used <-- correct

Answer: D. Keeping a separate validation partition is what allows the test set to be consulted only once, so its estimate stays honest.

Q7. Explain why the test partition should be consulted only once.

- A) Repeated use turns it into part of the development data <-- correct
 - B) Repeated use damages the stored data file
 - C) Repeated use makes the model train more slowly
 - D) Repeated use changes the model's parameters
- Answer: A.** Each time a choice is made on the strength of the test score, information from that set leaks into the model and the estimate becomes optimistic.

Q8. Distinguish a Pandas DataFrame from a NumPy array.

- A) A NumPy array carries column names and mixed types
- B) Both of the two carry column names and mixed types
- C) A DataFrame carries column names and mixed types <-- correct
- D) Neither of the two can hold numeric values

Answer: C. A NumPy array holds one data type and is addressed by position. A DataFrame labels its columns and can mix numbers, text and dates.

Q9. Describe what a correlation close to zero between two features indicates.

- A) No relationship of any kind between the two
- B) The two features are identical to each other
- C) No straight-line relationship between the two <-- correct
- D) One feature causes the other to change

Answer: C. A curved relationship can give a correlation near zero, which is why a scatter plot should be inspected as well as the coefficient.

Q10. Interpret a column in which 70 per cent of the entries are missing.

- A) The column is usually dropped rather than imputed <-- correct
- B) The column should be filled with its mean value
- C) The rows with a value should all be deleted
- D) The column is the most informative in the table

Answer: A. Imputing seventy per cent of a column invents most of it. If the fact of being present is itself informative, an indicator column can be kept instead.

Q11. Predict the effect of applying one-hot encoding to a column holding five thousand distinct categories.

- A) The column is reduced to a single column

- B) The categories are automatically merged
- C) The encoding fails and returns an error
- D) Five thousand sparse columns are created <-- correct

Answer: D. High-cardinality columns are usually grouped, or replaced by a frequency or target-based encoding, rather than expanded directly.

Q12. Recognize the plot described: vertical bars showing how many records fall into each category.

- A) A histogram
- B) A bar chart <-- correct
- C) A scatter plot
- D) A box plot

Answer: B. A bar chart counts categories; a histogram bins a continuous variable. The two look alike and are frequently confused.

Q13. Explain why a tree-based model does not need its features placed on a common scale.

- A) It converts every column to the same units
- B) It ignores every column except the largest
- C) It computes distances between the records
- D) It splits on thresholds within a single column <-- correct

Answer: D. Each split compares values inside one feature, so the units of the other features are irrelevant. Distance-based methods are the ones that need scaling.

Q14. Summarize what exploratory data analysis is carried out for.

- A) To fit the final model to the training data
- B) To report the accuracy of the finished system
- C) To learn what is in the data before any model is chosen <-- correct
- D) To deploy the model into the live product

Answer: C. Ranges, missing values, skew, outliers, class balance and correlations all decide what preparation and which model are appropriate.

Q15. Distinguish structured data from semi-structured data.

- A) Semi-structured data has fixed columns and no tags
- B) Structured data has fixed columns, semi-structured carries its own tags <-- correct
- C) Both of the two have a fixed set of columns
- D) Neither of the two can be stored in a file

Answer: B. A CSV table is structured; a JSON document is semi-structured, since its fields are named in the data itself and can vary between records.

Q16. Explain why an indicator column is sometimes added when missing values are filled in.

- A) The fact that a value was absent may itself be informative <-- correct
- B) The indicator improves the speed of training

- C) The indicator is required by the Pandas library
- D) The indicator replaces the need for imputation

Answer: A. A missing income may say something about the record. Keeping a flag lets the model use that, rather than treating the filled value as if it were observed.

Q17. Predict what happens to the number of rows when duplicate records are removed.

- A) It rises by the number of duplicates found
- B) It falls by the number of duplicates removed <-- correct
- C) It stays exactly the same as before
- D) It falls to the number of distinct columns

Answer: B. Duplicates give some records more weight than they deserve, and if they straddle a split they also leak information between the partitions.

Q18. Describe what stratified k-fold cross validation preserves.

- A) The proportion of each class within every fold <-- correct
- B) The original order of the rows in the table
- C) The exact number of features in every fold
- D) The mean of every numeric column per fold

Answer: A. With a rare class, an ordinary split can leave a fold with very few of its rows or none at all, making the score for that class unstable.

Q19. Explain why a logarithmic transformation is applied to a heavily right-skewed column.

- A) It removes every missing value from the column
- B) It converts the column into a categorical one
- C) It guarantees the column becomes normal
- D) It compresses the long tail and evens out the spread <-- correct

Answer: D. Income and transaction amounts are typically right-skewed, and the transformation stops a handful of very large values dominating what follows.

Q20. Distinguish filter feature selection from wrapper feature selection.

- A) A wrapper scores features alone, a filter tests them
- B) Both of the two test the features inside a model
- C) A filter scores features alone, a wrapper tests them in a model <-- correct
- D) Both of the two score the features in isolation

Answer: C. Filters are cheap and model-independent; wrappers are far more expensive but account for how features behave together in the model actually being used.

Q21. Apply 10-fold cross validation to a dataset of 1,500 rows and determine the number of rows in the validation partition of each fold.

- A) 150 <-- correct

- B) 1,350
- C) 15
- D) 1,500

Answer: A. The data is split into 10 equal folds, so each holds $1,500 / 10 = 150$ rows for validation and the remaining 1,350 are used for training.

Q22. Solve the following: a dataset of 2,000 rows is split with a test size of 0.2. Determine the number of training rows.

- A) 400
- B) 1,800
- C) 1,600 <-- correct
- D) 200

Answer: C. Twenty per cent of 2,000 is 400 test rows, leaving $2,000 - 400 = 1,600$ for training.

Q23. Use the interquartile range rule with a multiplier of 1.5 on a column whose first quartile is 20 and third quartile is 60, and determine the upper fence.

- A) 100
- B) 120 <-- correct
- C) 90
- D) 140

Answer: B. The interquartile range is $60 - 20 = 40$, so the upper fence is $60 + 1.5 \times 40 = 60 + 60 = 120$.

Q24. Compute the standardised value of an observation equal to 88 in a column whose mean is 70 and standard deviation is 12.

- A) -1.5
- B) 0.75
- C) 2.0
- D) 1.5 <-- correct

Answer: D. The z-score is $(88 - 70) / 12 = 18 / 12 = 1.5$, so the value lies one and a half standard deviations above the mean.

Q25. Apply min-max scaling to the value 45 in a column whose minimum is 5 and maximum is 105.

- A) 0.45
- B) 0.5
- C) 0.4 <-- correct
- D) 0.35

Answer: C. Min-max scaling gives $(45 - 5) / (105 - 5) = 40 / 100 = 0.4$.

Q26. Examine the following and determine the fault: a categorical column holding the cities Delhi, Mumbai and Chennai is encoded as 0, 1 and 2 and fed to a linear model.

- A) The encoding produces too many extra columns
- B) The encoding removes the missing values silently
- C) The encoding requires the column to be scaled

D) An order and a spacing have been imposed that do not exist <-- correct

Answer: D. A linear model will read Chennai as twice Mumbai. Unordered categories must be one-hot encoded; ordinal encoding suits only genuinely ordered categories.

Q27. Solve the following: a column of 800 entries has 120 missing values. Determine the percentage that is missing.

- A) 15 per cent <-- correct
- B) 12 per cent
- C) 20 per cent
- D) 8 per cent

Answer: A. $120 / 800 = 0.15$, so 15 per cent of the column is absent, which is normally low enough for imputation to be reasonable.

Q28. Apply the correct order of steps when a numeric column must be scaled before a model is trained.

- A) Fit the scaler on all rows, then split into two parts
- B) Split first, fit the scaler on the training rows, transform both <-- correct
- C) Split first, fit a separate scaler on each partition
- D) Transform both partitions before any scaler is fitted

Answer: B. Fitting on the whole table lets the mean and spread of the test rows influence the training data, and the reported score then comes out optimistic.

Q29. Use Pandas to determine the method that replaces missing entries with a stated value.

- A) dropna()
- B) isnull()
- C) describe()
- D) fillna() <-- correct

Answer: D. fillna() substitutes a value; dropna() removes the affected rows or columns; isnull() only reports where the gaps are.

Q30. Compute the number of columns produced by one-hot encoding a four-category column when no category is dropped.

- A) 3
- B) 4 <-- correct
- C) 1
- D) 16

Answer: B. One indicator column is created for each category. Dropping one of them avoids the redundancy that matters for a linear model.

Q31. Apply 5-fold cross validation and determine how many times the model is fitted.

- A) 1
- B) 10
- C) 5 <-- correct
- D) 25

Answer: C. Each fold in turn is held out for validation while the model is fitted on the remaining four, so five fits are performed in total.

Q32. Examine a scatter plot of two numeric columns showing a clear upward straight-line trend, and determine what it indicates.

- A) A strong positive linear relationship <-- correct
- B) A strong negative linear relationship
- C) No relationship between the columns
- D) A curved relationship between them

Answer: A. The scatter plot is the standard check on a correlation coefficient, since the same coefficient can arise from very different shapes.

Q33. Use the shape attribute on a DataFrame holding 500 rows and 12 columns and determine what it returns.

- A) (500, 12) <-- correct
- B) (12, 500)
- C) 500
- D) 512

Answer: A. The shape is reported as rows followed by columns, which is the first check to make after a table is loaded.

Q34. Solve the following: a table of 1,000 rows has 45 duplicates and 30 rows with a missing target removed, with no overlap between the two groups. Determine the rows that remain.

- A) 955
- B) 970
- C) 875
- D) 925 <-- correct

Answer: D. $1,000 - 45 - 30 = 925$ rows survive both cleaning steps.

Q35. Apply feature engineering to a column of order timestamps and determine a useful derived feature.

- A) The raw timestamp converted to a text string
- B) The hour of the day and the day of the week <-- correct
- C) The row number at which the order appears
- D) The number of characters in the timestamp

Answer: B. A model cannot see that 23:50 and 00:10 are close, or that Saturdays behave differently, unless those quantities are extracted into their own columns.

Q36. Compare a histogram with a box plot on what each one reveals about a single numeric column.

- A) The box plot shows the shape, the histogram the quartiles
- B) Both of the two show only the mean of the column
- C) The histogram shows the shape, the box plot the quartiles and outliers <-- correct
- D) Both of the two show the relationship between columns

Answer: C. A histogram exposes a second peak that a box plot conceals; a box plot marks candidate outliers that a histogram only hints at.

Q37. Use a variance threshold for feature selection and determine which feature it removes.

- A) The one with the largest range of values
 - B) The one most strongly correlated with the target
 - C) The one whose value is nearly the same in every row <-- correct
 - D) The one with the greatest number of categories
- Answer: C.** A near-constant column cannot distinguish between records, so it can carry no information whatever the model that follows.

Q38. Examine a training set in which one class holds only 2 per cent of the rows and determine the appropriate splitting strategy.

- A) A purely random split
 - B) A stratified split <-- correct
 - C) A split ordered by the target
 - D) No split, using all rows for training
- Answer: B.** A random split can leave the rare class barely represented, or absent, in one partition, making the score for that class unstable or impossible to compute.

Q39. Apply the train, validation and test arrangement and determine which partition is used to choose the settings.

- A) The training partition
 - B) The test partition
 - C) All three of the partitions
 - D) The validation partition <-- correct
- Answer: D.** Training fits the parameters, validation chooses among models and settings, and the test set is kept back for a single honest estimate at the end.

Q40. Relate the size of the test partition to the reliability of the estimate it produces.

- A) A larger test set gives a steadier estimate but less training data <-- correct
 - B) A larger test set gives a less steady estimate
 - C) The size of the test set does not affect the estimate
 - D) A smaller test set always gives a better estimate
- Answer: A.** The exchange is why cross validation is preferred on a small dataset: every row serves in turn for training and for validation.

----- End of Sample Question Set -----